

Component & Credentials List

AI Recruitment Automation — Day 32 Solution Architecture — Petalnex Pvt. Ltd.

1. Node Groups & Components

Component	Type / Integration	Purpose
Form Trigger	n8n node	Collects candidate applications (name, email, role, experience, skills, resume/cover text)
Field Validation	n8n If node	Confirms required fields are present before processing continues
Skills Extraction & Scoring	Google Gemini (LLM chain)	Extracts skills from candidate data and scores fit against role requirements
Output Parsing	n8n Code node	Parses and validates the structured output returned by Gemini
Rule-Based Assessment	n8n Code / If node	Applies minimum skill-match threshold before recruiter review
Candidate Data Store	Supabase (Postgres)	Stores raw submissions, processed candidate records, scores, and status history
Recruiter Notification	Gmail SMTP	Sends candidate summary email to the recruiter for review
Human Approval Gate	n8n Wait node	Pauses workflow until recruiter provides a decision
Decision Routing	n8n Switch node	Routes candidate to Shortlist, Reject, or Hold based on recruiter input
Candidate Notification	Gmail SMTP	Sends outcome email to the candidate (shortlist / reject / hold)
Final Status Update	Supabase (Postgres)	Records final candidate status and timestamp for audit trail

2. Credentials

Service	Credential Type	Notes
Supabase Postgres	Postgres credential (host, port, database, user, password, SSL mode)	Stored in n8n credential store; direct or pooler connection per environment
Google Gemini API	API key credential	Used for the LLM chain node(s); scoped to Gemini API access only
Gmail SMTP	App password credential (ID: Dx1F4PbUgZupfNSX, "SMTP account")	App-password auth avoids full OAuth2/Google Cloud Console setup
n8n instance	Self-hosted, Windows PowerShell	Access restricted to local/authorized network; workflows not publicly exposed

3. Failure Handling

Failure Scenario	Handling Approach
Missing required fields	If node blocks progression; submission logged, not silently dropped
Gemini API failure or timeout	Retry logic in Code node; structured failure object returned instead of throwing
Malformed LLM output	Parsing/validation Code node checks structure before downstream use
Email delivery failure	Logged for manual follow-up; does not block Postgres status update
Supabase connection failure	Credential validated at setup; SSL mode and pooler/direct connection selected per environment

4. Security Considerations

Area	Approach
Credential storage	All API keys and passwords stored in n8n's built-in credential store, never hardcoded in nodes
Data minimization	Only fields required for evaluation are stored; no unnecessary personal data retained
Human-in-the-loop	No candidate is finally rejected or shortlisted without explicit recruiter approval
Access control	n8n instance and Supabase project restricted to authorized internship/recruiter accounts
Audit trail	Every status change is timestamped and stored in Postgres for traceability